

Name _____

1km=1000m 1hour=3500s

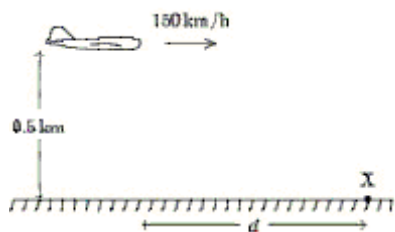
- 1) Suppose that a car traveling to the west (the $-x$ direction) begins to slow down as it approaches a traffic light. Which statement concerning its acceleration in the x direction is correct? 1) _____
 A) Its acceleration is negative but its velocity is positive.
 B) Its acceleration is positive but its velocity is negative.
 C) Both its acceleration and its velocity are positive.
 D) Both its acceleration and its velocity are negative.

- 2) Two projectiles of different mass are thrown in the air. Which one has a greater acceleration? 2) _____
 A) they have the same acceleration
 B) the one with the smaller mass
 C) we need to know the masses
 D) the one with the larger mass

- 3) As part of an exercise program, a woman walks south at a speed of 2.00 m/s for 60.0 minutes. She then turns around and walks north a distance 3000 m in 25.0 minutes (two answers) 3) _____
 (a) What is the woman's average velocity during her entire motion?
 A) 0.824 m/s south
 B) 1.93 m/s south
 C) 2.00 m/s south
 D) 1.79 m/s south
 E) 800 m/s south
 (b) What is the woman's average speed during her entire motion?
 A) 0.824 m/s
 B) 1.93 m/s
 C) 2.00 m/s
 D) 1.79 m/s
 E) 800 m/s

- 4) The position of an object as a function of time is given by $x = bt^2 - ct$, where $b = 2.0 \text{ m/s}^2$ and $c = 6.7 \text{ m/s}$, and x and t are in SI units. What is the instantaneous velocity of the object when $t = 2.2$? 4) _____
 A) 2.7 m/s
 B) 2.3 m/s
 C) 2.1 m/s
 D) 1.7 m/s

- 5) The plane flies at a constant speed 150km/h as shown. (0.5km above the ground). It releases a package at $t=0$. How far (horizontal) does the package travel during it's launch? 5) _____
 (take $g=10\text{m/s}^2$. hint: $1\text{km/h} = 0.28\text{m/s}$)

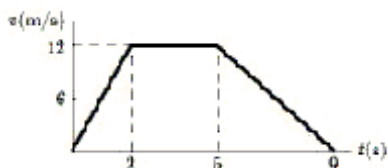


- A) 295m
 B) 150m
 C) 150m
 D) 420m

- 6) The position of an object is given as a function of time by $x=7t - 3t^2$, where x is in meters and t is in seconds. Its average velocity over the interval from $t=0$ to $t=4$ is: 6) _____
 A) -14..5
 B) -5
 C) 5
 D) -11

7) The graph represents the straight line motion of a car. between 5 and 9 s the car is

7) _____



On the y-axis you have $v(\text{m/s})$

- A) slowing down at a rate of 3 ms/s
 B) moving at a constant velocity of 3m/s
 C) slowing down at a rate of 12 m/s/s
 D) speeding up at a rate of 3ms/s

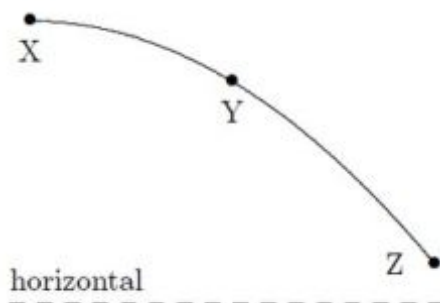
8) Refer to question 7. Between 2 and 5 s the car

8) _____

- A) is moving at a constant velocity
 B) is speeding up
 C) is not moving
 D) is slowing down

9) A stone is thrown horizontally and follows the path XYZ shown. The direction of the acceleration of the stone at point Y is

9) _____



- A) up
 B) down and right
 C) right
 D) down

10) A ball is thrown straight up with a speed of 36 m/s. How long does it take to return to its starting point, assuming negligible air resistance?

10) _____

- A) 7.3 s
 B) 11 s
 C) 3.7 s
 D) 15 s

11) An object moves 15.0 m north and then 11.0 m south. Find both the distance it has traveled and the magnitude of its displacement.

11) _____

- A) 26.0 m, 4.0 m
 B) 4.0 m, 4.0 m
 C) 4.0 m, 26.0 m
 D) 26.0 m, 26.0 m

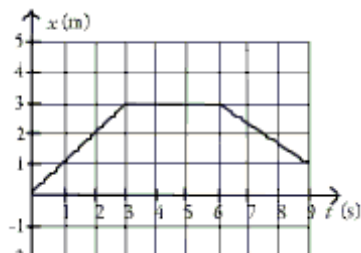
12) A car accelerates from 10.0 m/s to 30.0 m/s at a rate of 3.00 m/s^2 . How far does the car travel while accelerating?

12) _____

- A) 80.0 m
 B) 399 m
 C) 133 m
 D) 226 m

- 13) The figure shows the position of an object as a function of time, with all numbers accurate to two significant figures. Between time $t = 6$ s and time $t = 9.0$ s, the car

13) _____



- A) moving at constant speed (1m/s) @ right
B) moving at constant speed (1m/s) @ left
C) slowing down @ left (1m/s/s)
D) speeding up @ left (1 m/s/s)

- 14) A shell is launched with a velocity of 100 m/s at an angle of 30.0° above horizontal from a point on a cliff 50.0 m above a level plain below. How far from the base of the cliff does the shell strike the ground? There is no appreciable air resistance, and $g = 9.80 \text{ m/s}^2$ at the location of the cliff.

14) _____

- 15) A Monkey holds a coconut in a tree. A Coconut hunter tries to hit the coconut with an arrow. As soon as the hunter pulls the trigger, the Monkey lets go with the coconut. For a strike, the hunter should aim at

15) _____

- A) below the coconut
B) it does not matter
C) above the coconut
D) the coconut

- 16) A cat leaps to try to catch a bird. If the cat's jump was at 60° off the ground and its initial velocity was 2.74 m/s, what is the highest point of its trajectory, neglecting air resistance?

16) _____

- A) 0.19 m
B) 0.58 m
C) 10.96 m
D) 0.29 m

- 17) A car starts from rest and accelerates with a constant acceleration of 1.00 m/s^2 for 3.00 s. The car continues for 5.00 s at constant velocity. How far has the car traveled from its starting point?

17) _____

- A) 15.0 m
B) 4.50 m
C) 9.00 m
D) 24.0 m
E) 19.5 m

- 18) Vector $\vec{A}$ has a magnitude 5.00 and points in a direction 40.0° clockwise from the negative y axis.

18) _____

What are the x and y components of vector $\vec{A}$.

- A) $A_x = -3.21$ and $A_y = -3.83$
B) $A_x = -3.21$ and $A_y = 3.83$
C) $A_x = 4.29$ and $A_y = 2.16$
D) $A_x = 3.83$ and $A_y = 3.21$
E) $A_x = 3.83$ and $A_y = -3.21$

- 19) Two automobiles are 150 kilometers apart and traveling toward each other. One automobile is moving at 60km/h and the other is moving at 40km/h mph. In how many hours will they meet? (no conversion)

19) _____

- A) 1.5 h
B) 2 h
C) 2.5h
D) 1.75 h

- 20) EXTRA CREDITS. Galileo studied motions of uniformly accelerated object using an apparatus called:

_____ and he measured time using a _____

21)

Kinematics in 1-D:

$$\Delta x = x_f - x_i, \quad \bar{v} = \frac{\Delta x}{\Delta t}, \quad v = \frac{dx}{dt}, \quad \bar{a} = \frac{\Delta v}{\Delta t}, \quad a = \frac{dv}{dt}$$

Constant acceleration in 1-D:

$$v = v_0 + at$$

$$x - x_0 = \left(\frac{v + v_0}{2} \right) t$$

$$x = x_0 + v_0 t + \frac{1}{2} at^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

Answer Key

Testname: EXAM101_22

- 1) B
- 2) A
- 3) (a) A (b) C
- 4) C
- 5) D
- 6) B
- 7) A
- 8) A
- 9) D
- 10) A
- 11) A
- 12) C
- 13) B
- 14) 963 m
- 15) D
- 16) D
- 17) E
- 18) A
- 19) A
- 20)
- 21)